

A BOARD For CREATING Your IoT PRODUCT FASTER



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Building your own IoT device is not as simple as it looks. The first hurdle is in putting the hardware together. Then comes the process of faithfully transmitting messages over the distance you need the signal to travel. Let us not forget the time it takes to develop your solution and the cost for it.

This innovation story introduces an evaluation board that helps you get your idea into a finished product quicker, and also lets you create an open access mesh network to send your data reliably up to 2km. The board in question is a Bluetooth Low Energy (BLE) and ANT based board by Atoll Solutions Pvt Ltd.

BLE-ANT, a useful combination

What makes this board special is the way it is put together, enabling faster time-to-market and low-power solutions. The board contains Raytac module with Nordic Semiconductor nRF51 chip, which offers support to BLE and ANT protocols.

Enabling mesh creation. Generally, when you communicate via Bluetooth, you can transfer data up to a maximum of 80m in open air, and about 18m in normal scenarios. With BLE-ANT combination, you can create a mesh network and connect up to 2km radius, with all devices connected to this mesh network.

When a device sends out a signal, it is communicated via the mesh, from one node to the other, until it reaches the end device. Thus, information is secure and a strong signal is received by the end user.

Low power consumption. A device designed using this board is mostly in sleep mode, awoken by signals only when it has work to do. The claim is that a coin-cell battery combined with this board's power management scheme will let you use this board for about six months on a single charge.

Easily connected to existing setup.

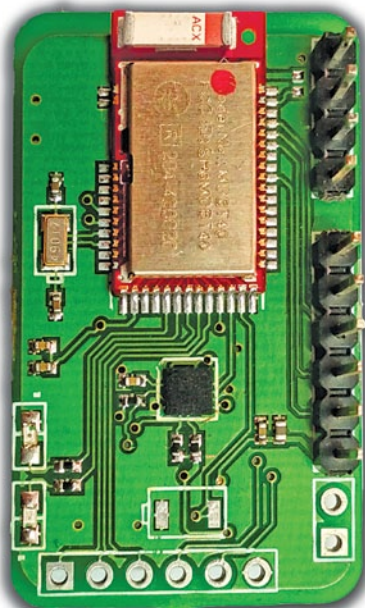
As soon as a module like ANT comes into the picture, the notion is that connecting to this might need a lot of effort from the user. This is where BLE module steps in, letting you connect to even your smartphone easily. Of course, there is some configuration you would need to do from your end, according to the application you choose, but the back-end is taken care of by the board.

Deep inside the board...

The board is built around Nordic Semiconductor nRF chip that comes in the form of MBDT40 board, with support for Bluetooth 4.0 and BT4.1 stack for BLE and ANT modules. It is designed based on Nordic nRF51422 solution. A 2.4GHz transceiver and an ARM Cortex 32-bit processor in the Nordic chip take care of the functioning.

Along with this, the board contains an accelerometer, ten configurable general-purpose inputs/outputs, three configurable analogue-to-digital converter channels, one 32-bit and two 16-bit timers with counter mode, and support for inter-

The evaluation board



Specifications of the board

Size: 18mm×10mm×3.2mm
Operating temperature: 0°C to 70°C
Operational relative humidity: 10 per cent to 80 per cent
Operating voltage: 1.8V to 3.6V
Frequency band: 2400MHz - 2483MHz
Supported data rates: 2Mbps, 1Mbps and 250kbps
Receiver sensitivity: -93dBm at BLE mode
Transmission power: -20dbm to +4dbm